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COLLECTING DUCT AND MICTURITION REFLEX

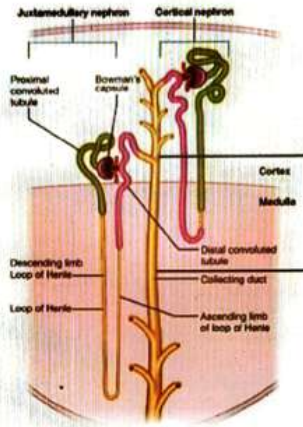
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Collecting Duct

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Hormones influencing collecting duct :

1. Aldosterone.
2. Anti-diuretic hormone/vasopressin.



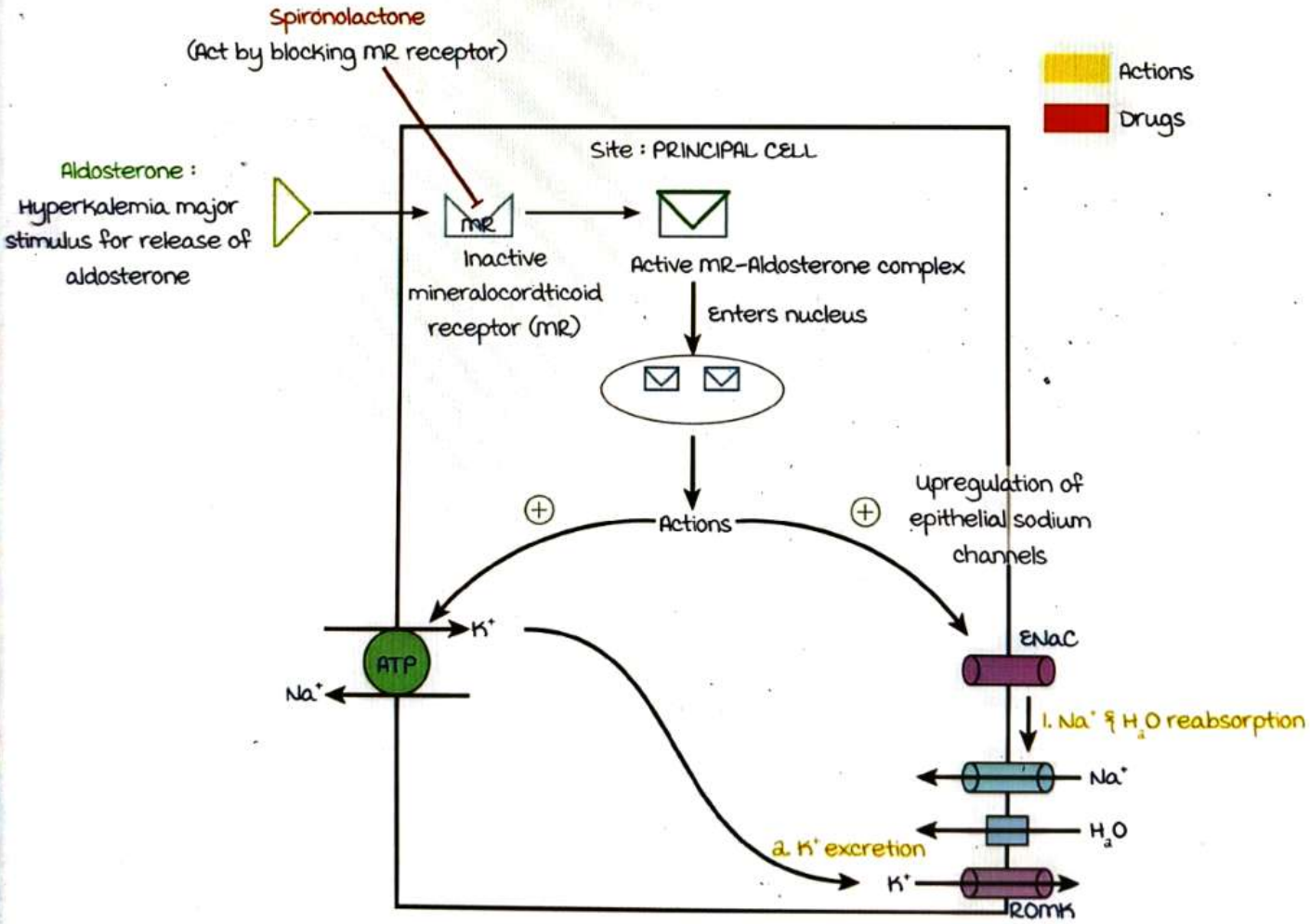
1. Cortical collecting duct (CCD)
2. Medullary collecting duct (MCD)

Parts of collecting duct

ALDOSTERONE :

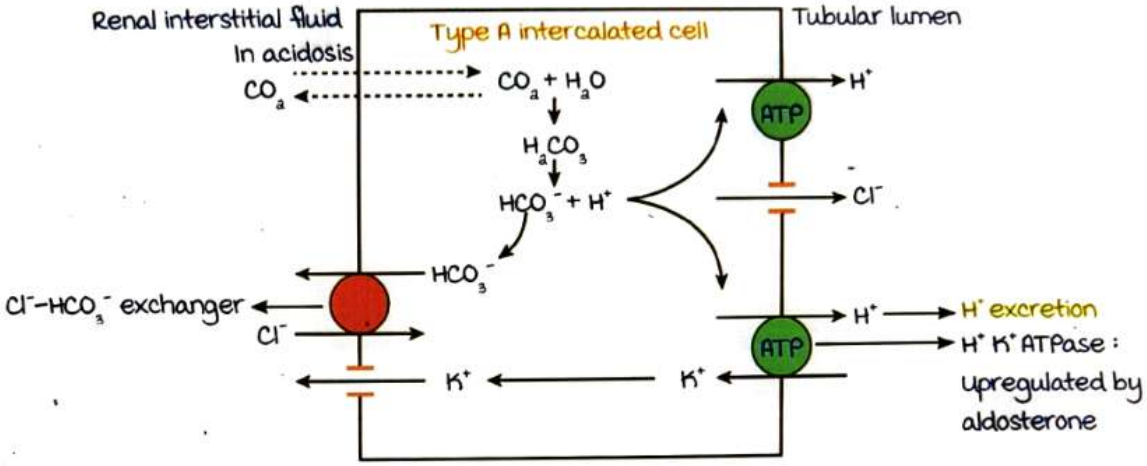
- Steroid hormone produced by adrenal cortex.
- Can cross the cell membrane.

mechanism of action :



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Applied aspect :

Amiloride : Blocks ENaC.

- Use : Diuretic (D/t Na^+ and H_2O excretion in the urine).

Spironolactone : Blocks MR.

- Use : K^+ sparing diuretic.

Aldosterone inactive → 1. Na^+, H_2O excretion.
2. K+ spared (Not excreted).

Liddle's syndrome : Overactive ENaC.

- ↑↑ Na^+ and H_2O reabsorption → Body fluid volume → Hypertension.

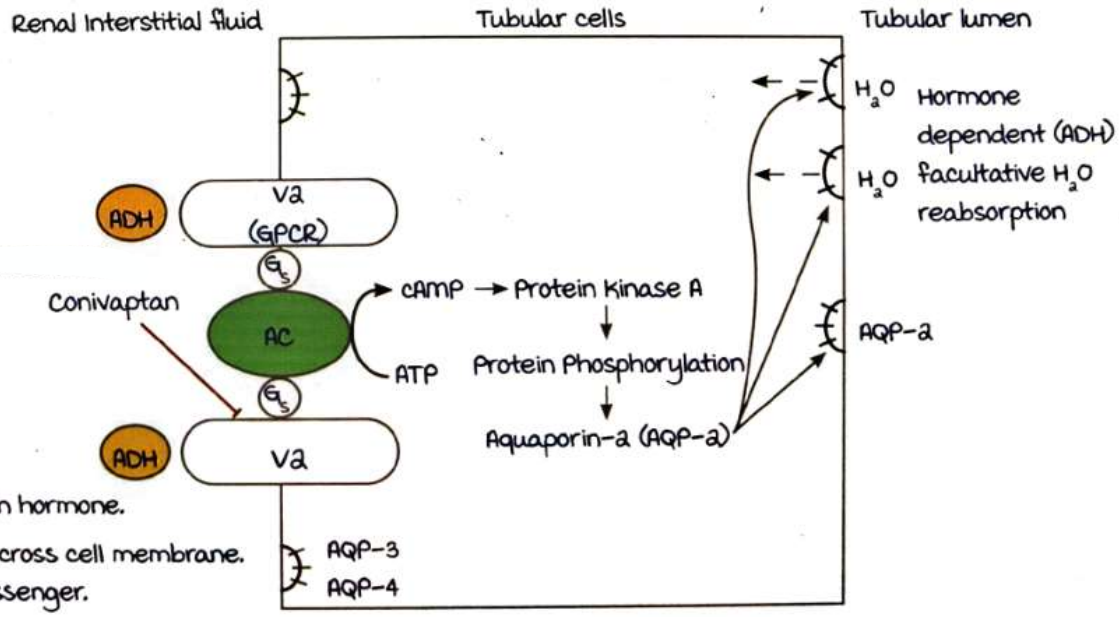
ADH/VASOPRESSIN :

Site of action in collecting duct :

1. P cells.
2. medullary collecting duct.

mechanism of action :

major action : Solute free water reabsorption by facilitated diffusion.



ADH :

- Protein hormone.
- Can't cross cell membrane.
- 1st messenger.

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Applied aspect :

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1. Aquaporins :
   - Vaptans (conivaptan) : V2 receptor blockers.
     - ↑ pure water excretion in urine.
2. Vasopressin in cancer : SIADH.
   - Kidney escapes the effect of ↑↑ ADH by downregulation of AqP-2.

Factors affecting ADH secretion :

↑ ADH Secretion : For water conservation.
- ↑ osmotic pressure.
- ↓ ECF.
- Painful stimuli.
- Nausea and vomiting.

↓ ADH Secretion : Water excretion.
- ↓ osmotic pressure.
- ↑ ECF volume.
- Alcohol (Inhibits ADH).

Actions mediated through Vasopressin receptors :

Receptor	Site	Action of ADH	ADH analogue & its use
V1	Blood vessel	Vasoconstriction (↓ blood loss in vascular injury)	Drug : Terlipressin. Use : Rx of esophageal varices.
V2	Collecting duct, vascular endothelium	Release of vWF from vascular endothelium	Drug : Desmopressin (ROA : Intranasal). Use : Von Willebrand disease.
V3	Anterior pituitary	ACTH release	-

Free water clearance :

Free water clearance	Concentration of urine	Level of ADH	Seen in
Negative	High	High	Syndrome of inappropriate ADH secretion (SIADH)
Positive	Low	Low/absent	Diabetes insipidus

- Free water clearance is zero → Kidney has lost the ability to dilute or concentrate urine.
- Max. water reabsorption : PCT (67 %).

Natriuretic Peptides

00:33:47

Types :

	A type (ANP)	B type (BNP)	C Type (CNP)
Site	Atrium	Cardiac ventricles	Vascular endothelium

Site :

Terminal DCT and CD.

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Action :

1. Natriuresis : ↑ Na^+ excretion.
   - Dilatation of afferent arteriole ⟶ ↑ GFR ⟶ ↑ Na+ filtration & excretion.
2. Physiological response regulating ANP :
   - Fluid overload, ↑ ECF (Excess Na^+ & H_2O) ⟶ ↑ ANP.
   - Hypovolemia, ⟶ ↓ central nervous pressure (low Na^+ & H_2O) ⟶ ↓ ANP.
3. Physiological antagonism ⟶ maintains homeostasis.
   - ↑ Na^+ : Activates natriuretic peptide.
   - ↓ Na^+ : Activates Renin-Angiotensin system (RAAS).
4. Aldosterone escape :

   Kidneys escape the effect of aldosterone in hyperaldosteronism.

   ↑↑ aldosterone ⟶ ↑ Na^+ & H_2O reabsorption ⟶ ↑ BP, no edema mechanism :

   a. ↑ BP ⟶ Pressure diuresis.

   b. Atrial natriuretic peptide (Aldosterone antagonist) : Causes natriuresis.

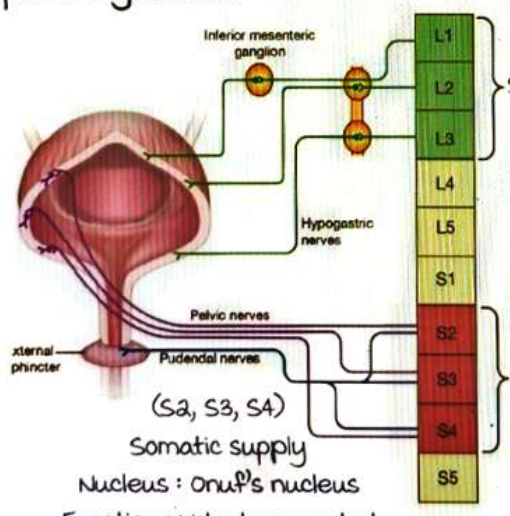
Micturition Reflex

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Smooth muscles in ureter —Pacemaker→ Peristalsis of urine ⟶ Bladder ⟶ micturition.

NEURAL CONTROL :

Spinal segments :



Sympathetic : mediates pain signals from bladder. Prevents retrograde ejaculation.

Parasympathetic : Facilitates micturition.

(S2, S3, S4)
Somatic supply
Nucleus : Onuf's nucleus
Function : Voluntary control

Higher centres :

Facilitatory centres : Facilitates micturition.
- Pons : Barrington centre.
- Posterior hypothalamus :

Inhibitory centre : Inhibits micturition.
- midbrain.

Stimulus : Cold
↓
↑ frequency of micturition

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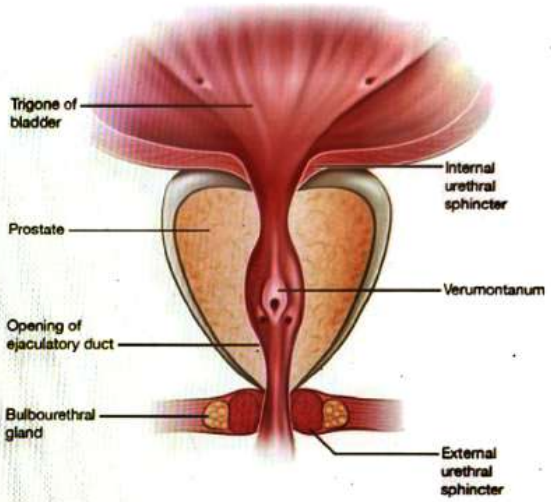
Centre for control of micturition :

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- Site : Paracentral lobule of cortex.
- Function : Highest order of control.
- Blood supply : Anterior cerebral artery.
- Lesion : Urinary incontinence.

REFLEX :

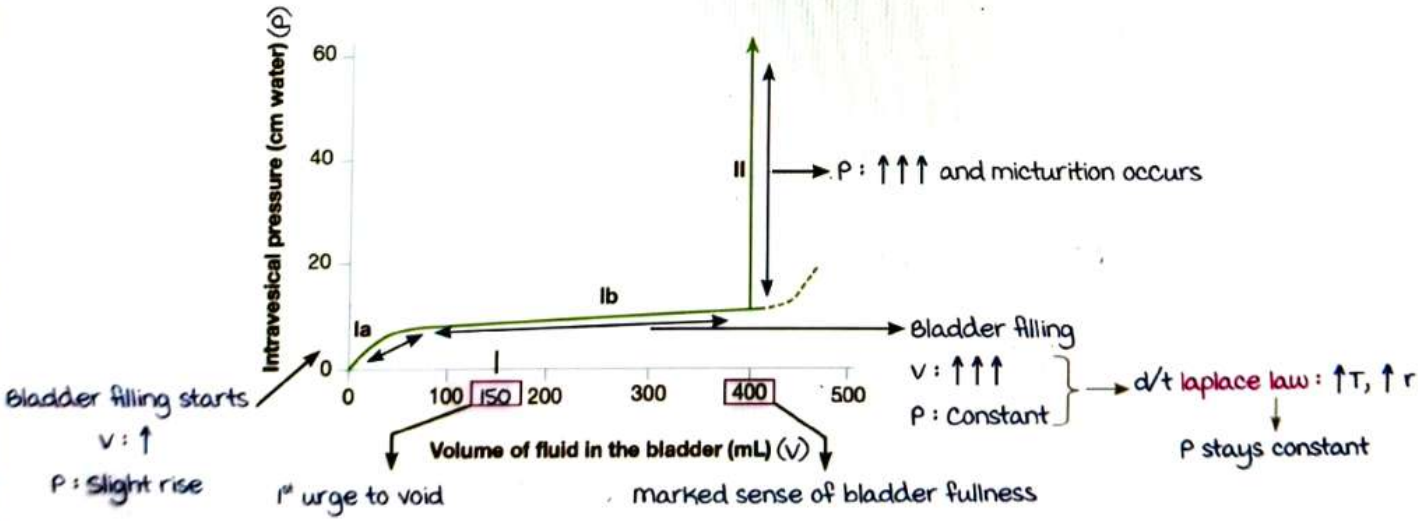
Bladder filling → Stretch
↓ Stimulus
Afferents to spinal cord
↓
Parasympathetic activation
↓
Contraction of detrusor
Relaxation of internal urethral sphincter
↓
Relaxation of external urethral sphincter → micturition



Int & ext urethral sphincter

CYSTOMETROGRAM :

Study of volume and pressure changes in the bladder.



Feedback